

Wrong stars inflate little red dot hosts

Field-star PSFs misassign light when source spectra differ, faking extended emission in compact high-redshift nuclei.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for little red dots and compact nuclei

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Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from nearby field stars. This assumes the target and the stars share the same spectral shape across the filter.

Nobody had quantified how much broad-band colour mismatch between source and reference stars biases inferred host flux.

What this paper is selling

The authors derive the physical and mathematical origin of this polychromatic PSF error and demonstrate a correction. Applied to little red dots, it changes inferred sizes, luminosities and masses.

Key results to remember

- A pure point source can yield an 11.8% fake F444W host component.
- The spurious allocation survives every tested nuclear spectrum, stellar reference and detector position.
- Source-conditioned PSFs are needed for reliable host measurements of compact distant objects.

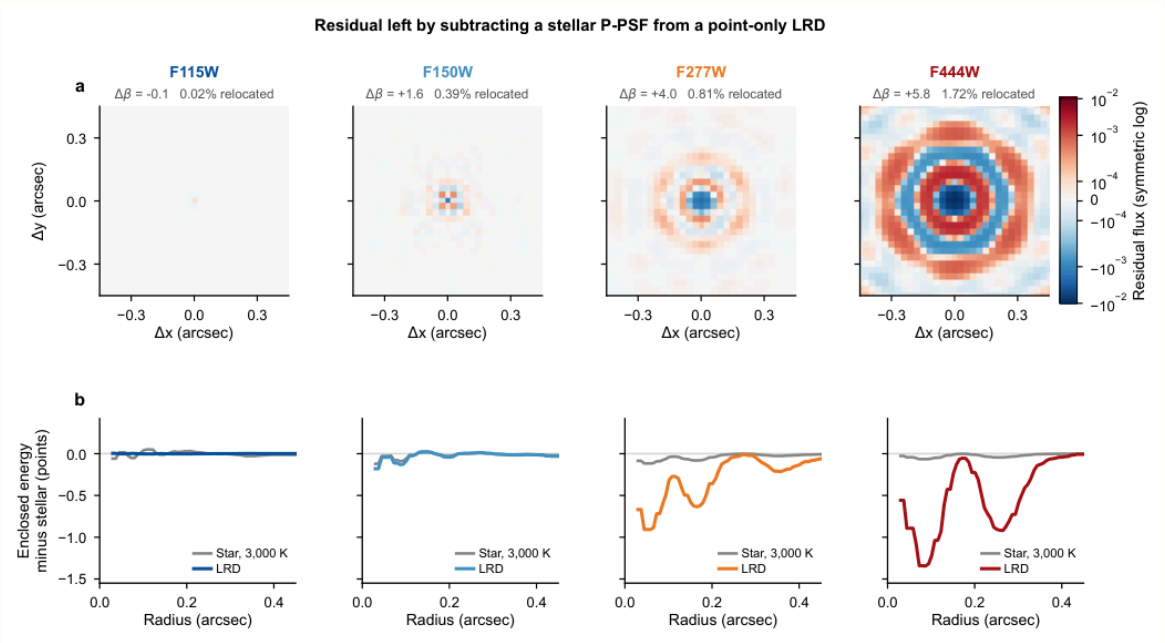


Fig. 1. Subtracting a stellar PSF from a point-only LRD leaves wavelength-dependent residuals that...

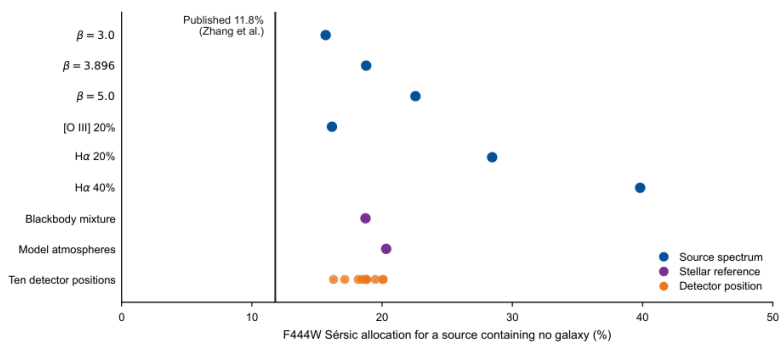


Fig. 2. Every tested spectrum, stellar reference and detector position allocates more F444W flux...